PARISHRAM 2.0

CLASS 12TH 2026

Relations and Functions

MATHS | LECTURE-4

BY-MANDAR SIR



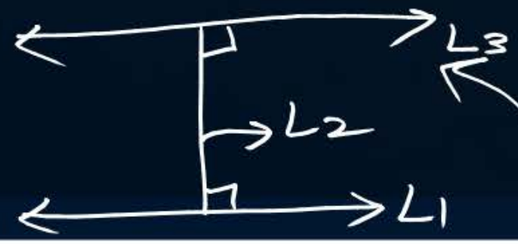
TOPICS TO BE COVERED

1. Questions on Relations (1 Question)
2. Introduction to Functions

Vapas Aaj Karenge Ekdum Basics se



QUESTION



Let L be the set of all lines in a plane and R be the relation in L defined as $R = \{(L_1, L_2) : L_1$ is perpendicular to $L_2\}$.
Show that R is symmetric but neither reflexive nor transitive.

(a,a)

Reflexive ✗

$R = \{(L_1, L_1) \; (L_1 \perp_r L_1)\}$

$L_1 \perp_r L_1$?

line 1

Line 1 is not $\perp_r$ to Line 1

$(a,b) \in R \quad (b,a) \in R$

Symmetric

$R = \{(L_1, L_2), L_1 \perp_r L_2)$

$R = \{(L_2, L_1), L_2 \perp_r L_1)$

↳ same

Symmetric ✓

Transitive

$(L_1, L_2) \in R \rightarrow L_1 \perp_r L_2$

$(L_2, L_3) \in R \rightarrow L_2 \perp_r L_3$

Check

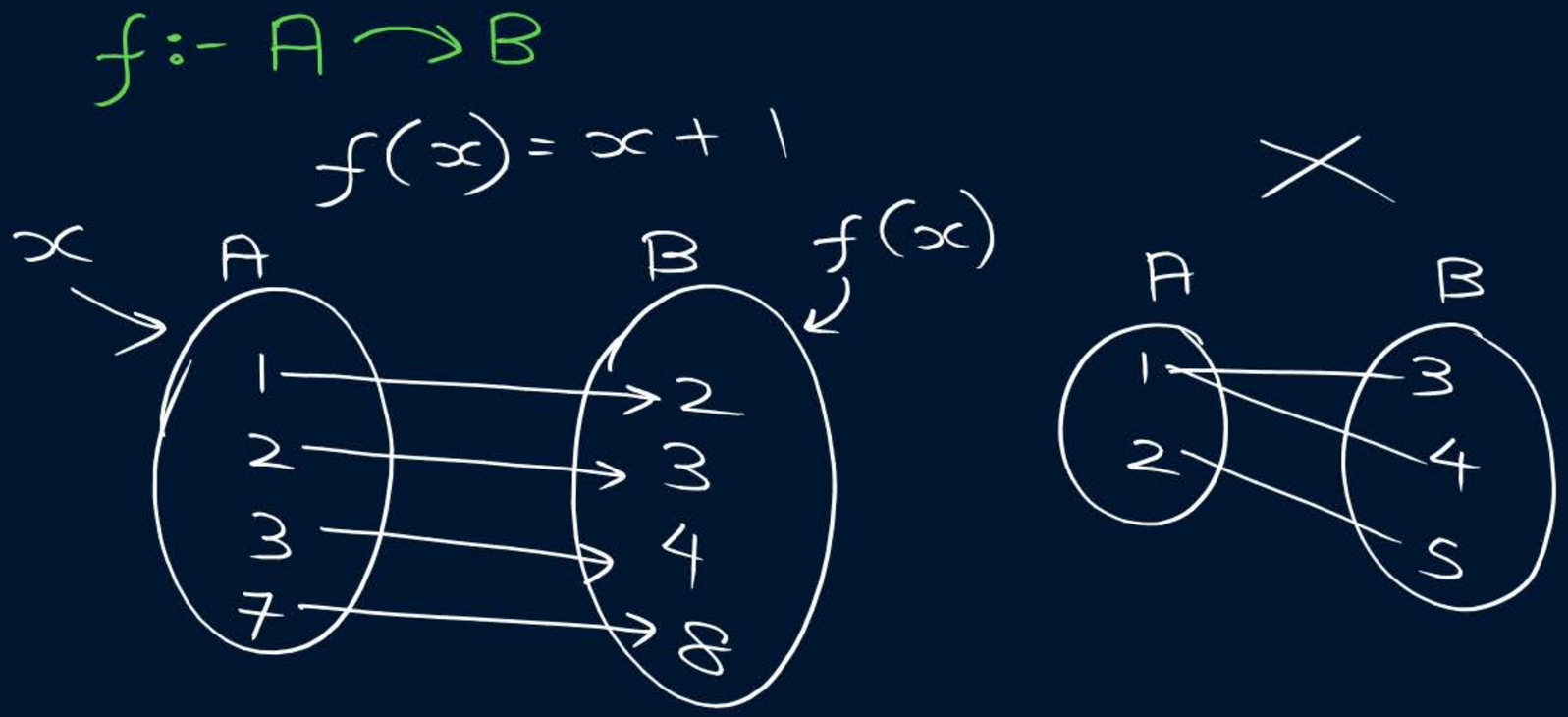
$(L_1, L_3) \notin R$

$L_1 \perp_r L_3$ ✗

Transitive ✗

What are Functions

A relation f from set A to a set B is said to be function if every element of set A has one and only one image in set B.



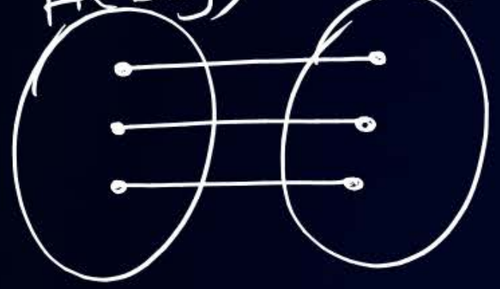
Features of Function f : A → B

1. For each element $x \in A$, there exist a unique element $y \in B$.
2. If there is an element in A which has more than one image in B, then $f : A \to B$ is not a function.
3. Different elements of set A may be associated with the same element of B.
4. If there is an element in A which does not have an image in B, then $f : A \to B$ is not a function.

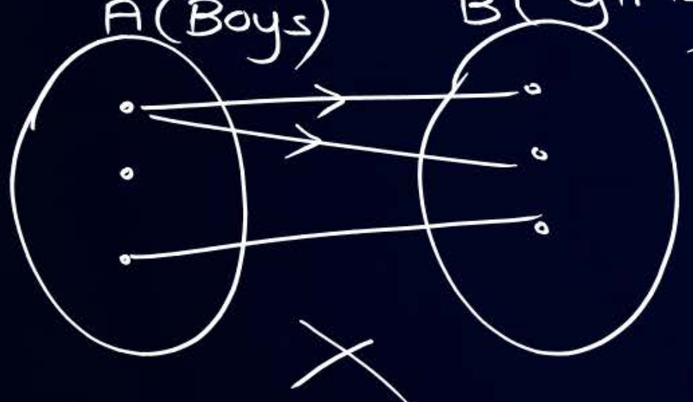
Mandar Style

A → Boys B → girls

1) $f: A \to B$

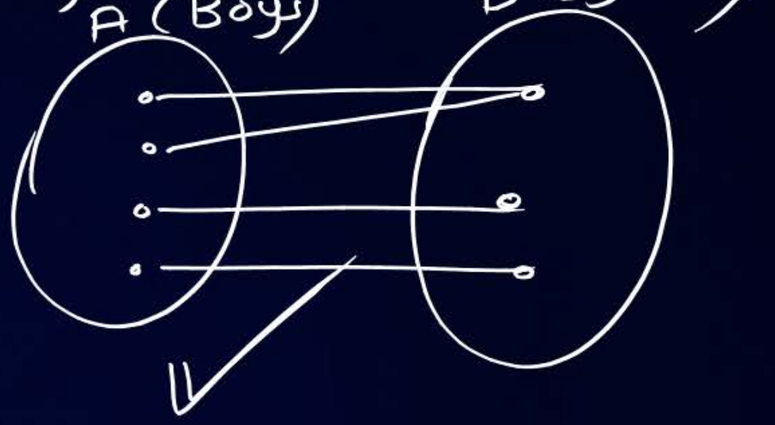


2) $f: A \to B$

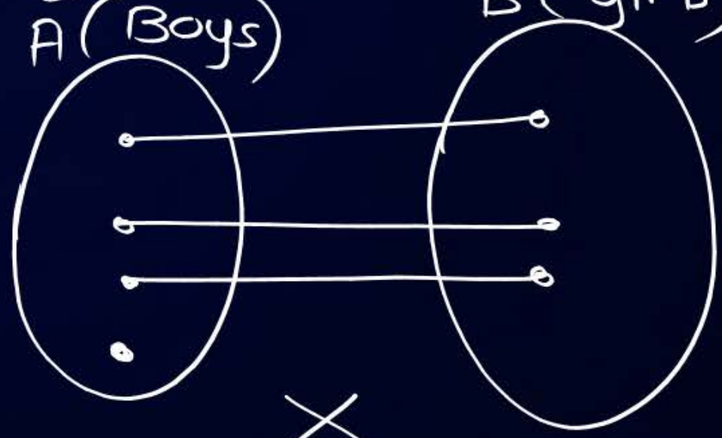


Not a function

3) $f: A \to B$

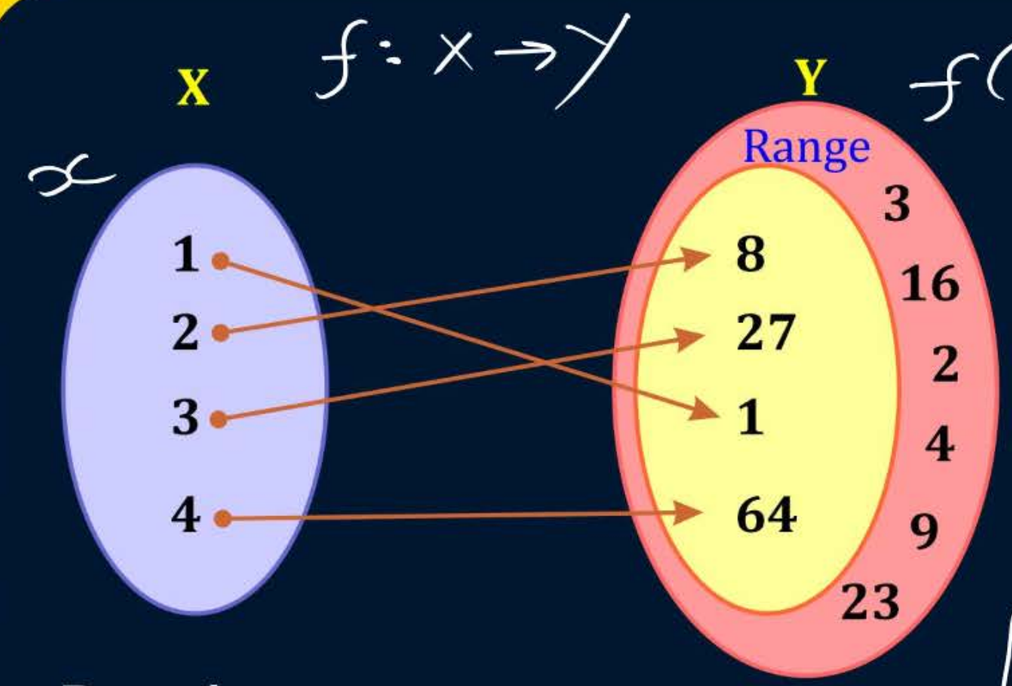


4) $f: A \to B$



Not a function

Domain, Range and Codomain of a Function



$f: X \to Y$

$f(x)$

Range
[Output Values]

$f(x) = x^3$

1	$f(1) = 1^3 =$	1
2	$f(2) = 2^3 =$	8
3	$f(3) = 3^3 =$	27
4	$f(4) = 4^3 =$	64

Domain [Input]

[Output] Range

$f: A \to B$

B saare elements

Co-domain

Domain, Range

$f(0) = 0$

$f(x) = x^2$

$f(1) = 1^2 = 1$

$f(2) = 2^2 = 4$

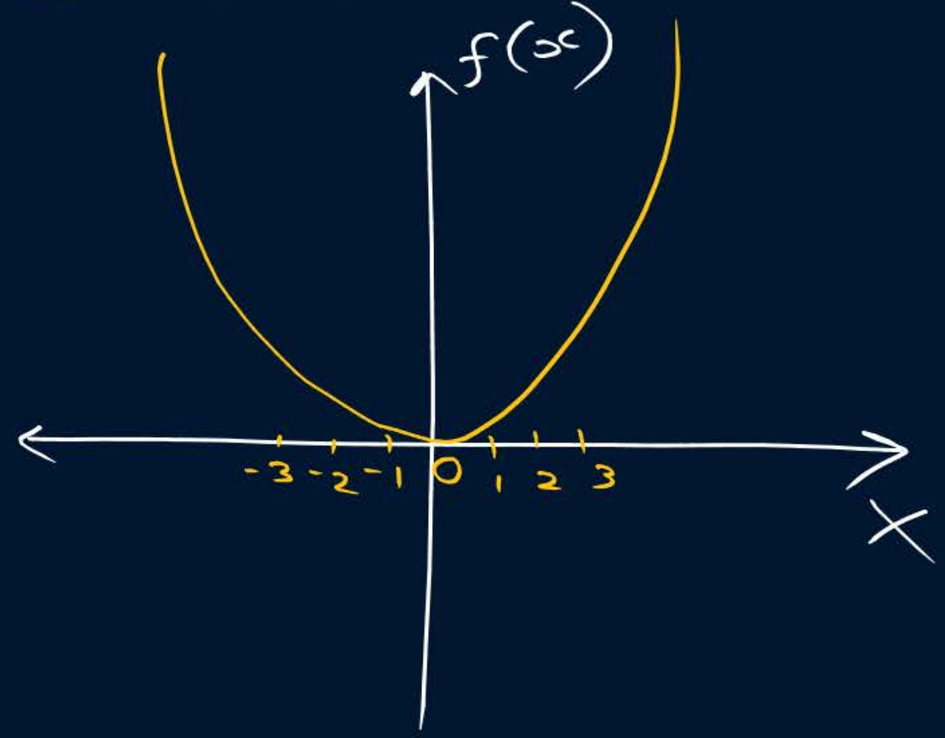
$f(3) = 3^2 = 9$

$f(-2) = (-2)^2 = 4$

$f(-3) = (-3)^2 = 9$

Domain $\rightarrow [-\infty, \infty]$

Range $\rightarrow [0, \infty)$

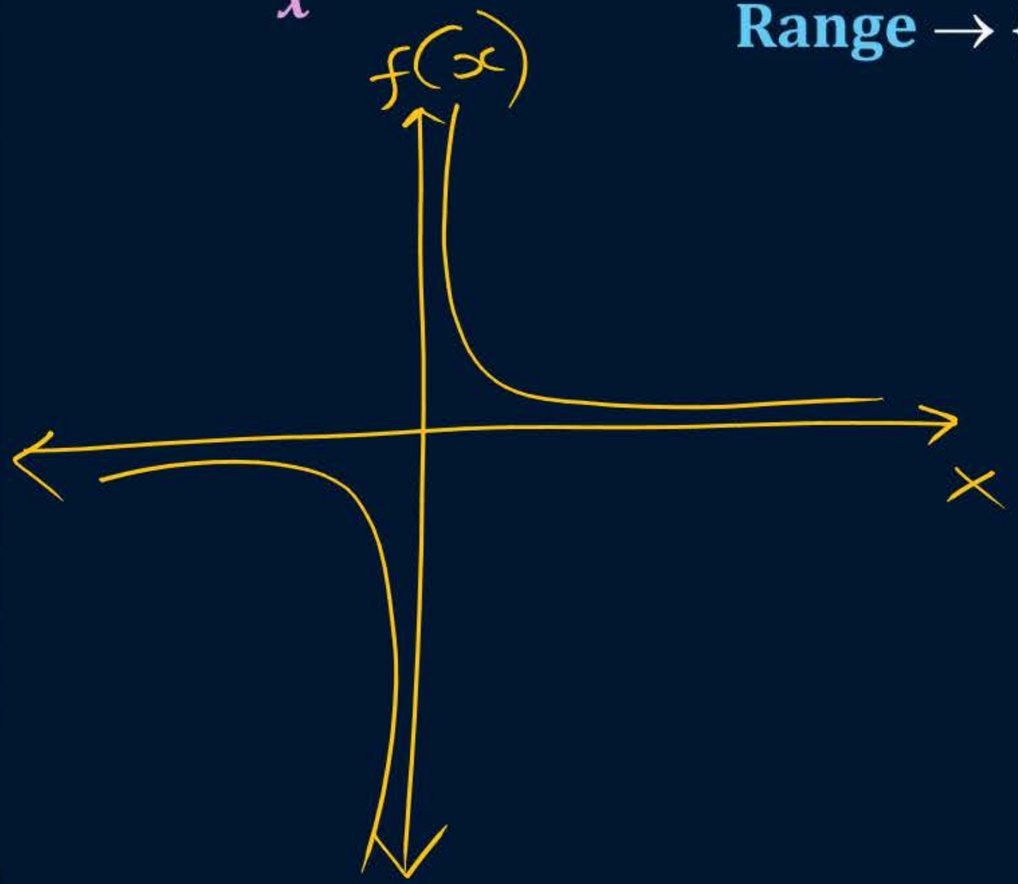


Domain, Range

$f(x) = \frac{1}{x}$

Domain $\rightarrow \{R - 0\}$

Range $\rightarrow \{R - 0\}$



$f(x) = \frac{1}{0}$ ✗

Denominator can't be zero

Domain, Range

$f(x) = \sqrt{x}$

Comment

Domain $\rightarrow [0, \infty)$

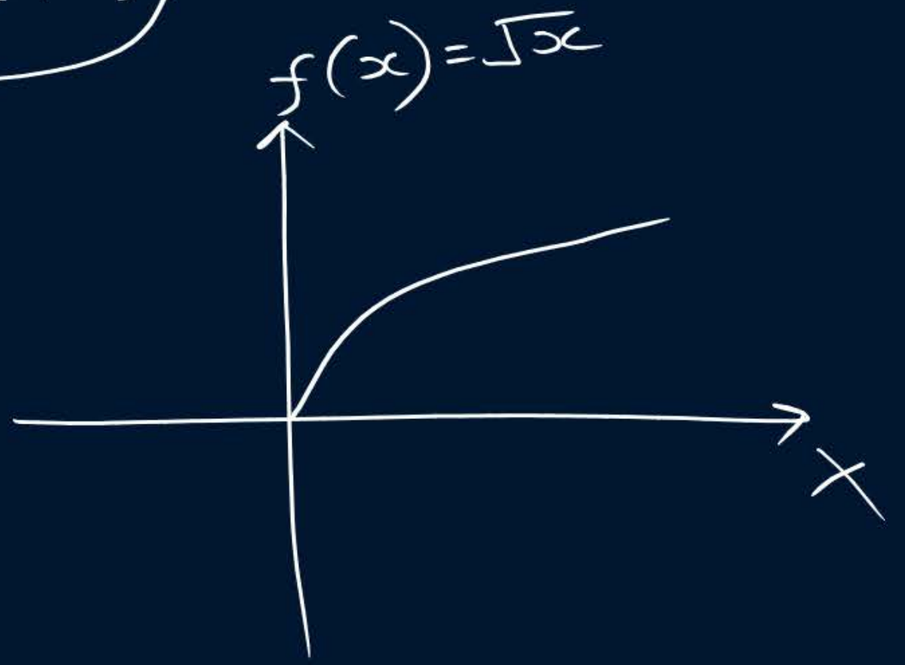
Range $\rightarrow [0, \infty)$

$\sqrt{+ve}$ → only positive values

input → only positive values

Domain → $[0, \infty)$

Range → $[0, \infty)$



Modulus function

$f(x) = |x|$

→ input app kuch bhi daalo

→ output → (+ ve)



$|-3| = +3$

$|+3| = +3$

THANK YOU